Cantera vs NN | Test IC 18 | NRMSE [%]

T0 = 1069.1 K P0 = 2.170 atm phi0 = 1.544 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	4.16	6	11.4	5.57	3.77	85.4	6.67	4.69	4.07	N/A	Complete
2 P-only DeepONet10	**10k**	10.9	8.4	11.6	6.12	8.64	36	8.86	7.08	8.04	N/A	Complete
2 P-only DeepONet10	**long**	1.2	2.1	2.65	2.17	1.91	2.31	1.79	2.15	2.28	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	4.73	5.19	4.52	4.39	5.19	13	5.06	6.03	6.35	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	1.93	2.94	3.57	3.07	2.71	3.71	2.59	3.2	3.25	N/A	Complete
4 T/P DeepONet10	**10k**	10.6	8.84	15.3	6.18	8.75	45.3	8.23	9.08	19.3	N/A	Complete
4 T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	**Stop 0.552 ms**
5 ANN10 + T/P DeepONet10	**10k**	11.1	7.78	14.8	5.93	7.79	44.7	7.91	7.43	7.5	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	60.5	13.7	104	2.5	1.45	238	15.4	1.5	1.97	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.250 ms**
6 T/P/ϕ DeepONet10	**long**	75.7	17.3	126	15.9	6.39	243	23.4	126	7.37e+04	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	49.7	9.18	49.4	3.65	9.37	374	26.3	9.37	19.7	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	63.5	11.7	103	7.32	0.634	258	17.9	1.14	1.02	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.